

VICTOR IZEKOR

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EDUCATION

University of Maryland, Baltimore County (UMBC)

Baltimore, MD

Bachelor of Science in Bioinformatics — Computer Science & Engineering Track

Expected May 2027

Relevant Coursework: Computer Science I & II, Data Structures & Algorithm Analysis (CMSC 341), Object-Oriented Programming, Statistics & Quantitative Analysis, Bioinformatics Algorithms

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, HTML/CSS — primary proficiency in Python and C++

Bioinformatics & Data: Sequence alignment algorithms (Needleman-Wunsch, Smith-Waterman), BLAST, UniProt, statistical data validation, data cleaning & visualization

Software Engineering: Data structures, object-oriented design, algorithm complexity analysis (Big-O), version control (Git), testing & debugging

Tools: Microsoft Office Suite, GCC/G++ compiler, command-line development environments

EXPERIENCE

Johns Hopkins Bayview Medical Center

Baltimore, MD

Patient Care Technician Intern — Neuro ICU

October 2025 – Present

- Collaborate daily with physicians, nurses, and specialists in the Neuro ICU, translating complex clinical and neurological data into clear documentation used across a multidisciplinary care team.
- Interpret and record neurological findings for patients with TBIs, hemorrhages, and tumors under time-sensitive, high-stakes conditions — building close familiarity with the clinical data and terminology used in oncology and neuro-critical care.

PROJECTS

Sequence Alignment Algorithms — Computational Bioinformatics

UMBC Bioinformatics Coursework

- Implemented Needleman-Wunsch (global) and Smith-Waterman (local) dynamic programming algorithms in Python for pairwise sequence comparison, evaluating accuracy and runtime trade-offs between alignment strategies.
- Queried and processed real biological datasets via BLAST and UniProt, applying statistical validation to algorithmic output — hands-on experience with tools used directly in genomics and cancer-research pipelines.

Custom Hash Table — Data Structures & Algorithm Design

UMBC CMSC 341

- Engineered a hash table in C++ using separate chaining, with dynamic resizing, collision resolution, and full manual memory management.
- Validated correctness with a dedicated testing framework across edge cases; performed Big-O analysis and tuned for $O(1)$ average-case lookup.

Cipher Encryption/Decryption Engine — Applied Software Design

UMBC CMSC 202

- Implemented three encryption algorithms (Atbash, Scytale, Omg) in C++ using a modular class hierarchy with clear interface contracts.
- Applied encapsulation, inheritance, and polymorphism to support new cipher modules without refactoring core logic.

Pulsify — Full-Stack Health-Tech Prototype

February 2025

Howard University Bison Bytes Hackathon

- Designed and built a full-stack web application (Python/Flask backend, JavaScript/HTML/CSS frontend) end-to-end within 24 hours, owning requirements, architecture, and key implementation decisions on an interdisciplinary team.

ADDITIONAL INFORMATION

Interests: Bioinformatics & computational biology, healthcare technology, cancer/genomic data analysis, applied machine learning, full-stack development